

Data Warehouses and Data Mining Systems

Laboratory 5: Report content

1 Analysis of data sources

Pay a special attention to datasets that have been chosen for your project. Try to find any problems (e.g. missing data, incompatible formats, NULL values, outliers etc.). In your report, describe what you have learned and try to provide a solution for these problems.

2 ETL approach

There are many approaches to performing an ETL process. All of them, however, share the same 3 stages: extraction, transformation and loading. First decide, what approach best suits your needs:

1. A dedicated ETL (or data integration) software/library/framework (e.g. SSIS) – it lets you define the dataflow (often via GUI) and connect any sources and data destinations;
2. A self-developed software (e.g. a Python script) – you are the architects of the ETL process and you design and implement a software that extracts, transforms and loads data into a data warehouse;
3. An ELT (Extract, Load & Transform) approach – in this approach you first load untransformed data into a database (other than your data warehouse) and then, using e.g. SQL queries, stored procedures and user-defined functions – you transform the data and put it into the data warehouse.

3 Data staging area

Data staging area is an additional database that is used to store source data that are going to be loaded into the data warehouse. It contains tables that has *the same* format/schema as the source data and is primarily used to gather all source data together in a one place. It is not always obligatory to use a data staging area, however in the ELT approach, it is part of the design process.

If you decide to use a data staging area, please include in your report information about its schema and technology used, as well as the queries/methods used to load data from sources into the staging area.

4 Transformations

Prior to loading data into a data warehouse, data need to be properly transformed, which includes e.g. cleaning, aggregating, changing formats, and all other operations that are required in order to have a consistent dataset for analysis. In your report, describe with details all the transformations that you are going to perform, basing on analysis made in section 1. Depending on the chosen approach (in section 2), describe all the technical details.

5 Loading phase

The last part of the ETL process is to load data into the data warehouse. This step requires you to perform some additional transformations, in order to split or join data – the schema of the data warehouse and data sources are different. In your report describe what operations were needed in this phase.